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2.1 - Narrow-Complex Tachycardia 1: Intro & pSVT SOAP

1) “100” next to heart timer:
Tachyarrhythmia = ventricular rate **>100 bpm (Narrow complex tachy/ Intro)**

2) Narrow beam over 3 boxes:
Narrow QRS complex = **<120 ms** (3 boxes) → indicates that impulse originated at/above AV node (supraventricular tachycardia; SVT) to His Purkinje system of ventricles **(Narrow complex tachy/ Intro)**

3) “Super event” around narrow beam:
Narrow QRS complex = **<120 ms** (3 boxes) → indicates that impulse originated at/above AV node (**supraventricular tachycardia; SVT**) to His Purkinje system of ventricles (**Narrow complex tachy/ Intro**)

4) Regular vs irregular notes:
after identifying a narrow complex (<120 ms QRS) tachycardia (>100 bpm), assess whether the rhythm is **regular** (sinus tach, AVRT/AVNRT, A flutter) vs **irregular** (A flutter, A fib, MAT) (**Narrow complex tachy/ Intro**)

5(O) “Sign-ups” with “P” tails before every QRS skirt:
sinus tachycardia (most common SVT) ⇒ SA node → conduction system of atria (upright P wave II, III, aVL, negative P wave aVR) → AV node → His Purkinje system (narrow QRS complex); **P wave precedes every QRS (Sinus tach/ ECG)**

6) Atrial dog signal:
supraventricular tachycardia (SVT) = any tachycardic rhythm that starts above the level of ventricle (e.g. sinus tach, a fib, a flutter) (**pSVT/ Intro**)

7) “Special access” “Reentry”:
paroxysmal supraventricular tachycardia (pSVT) = tachycardia caused by a reentrant circuit (e.g. AVRT = **accessory pathway**, AVNRT = **nodal reentrant circuit**) (**pSVT/ Intro**)

8(O) “Reentry 1:50-2:50” underneath regularly spaced notes:
pSVT (AVRT, AVNRT) = sudden/recurrent episodes of **regular tachycardia 150-250 bpm**(**pSVT/ ECG**)

9(S) “Episodes”:

pSVT (AVRT, AVNRT) = **sudden/recurrent episodes** of regular tachycardia 150-250 bpm (**pSVT/ HPI**)

10(O) Inverted “P” tail hiding behind QRS skirt:
pSVT (AVRT, AVNRT) → no discernable P waves before every QRS (impulse does not start at SA node) → inspect for **retrograde P waves** (buried inside QRS, inverted, appearing at the wrong time) (**pSVT/ ECG**)

11) Chasing tail within AV node:
Atrioventricular nodal reentrant tachycardia (AVNRT) = reentrant circuit **within** the AV node itself (**pSVT/ Intro**)

12) Extra path:
Atrioventricular reentrant tachycardia (AVRT) = accessory pathway between the atria and ventricle **outside** of the AV node (**pSVT/ Intro**)

13(O) “Ortho” path from ventricle to atria:
Orthodromic Atrioventricular reentrant tachycardia (AVRT) = impulse travels up accessory pathway (ventricle → atrium) → down through AV node to activate ventricles (**narrow QRS complex, regular ~150-250 bpm**) (**pSVT/ ECG**)

14(O) Dog with “fluttering” ears and intermittent jumps:
Atrial flutter = reentrant circuit or a single irritable area within atrium → atria ~300 bpm (**“sawtooth” flutter waves**) → some signals filtered through AV node to ventricles (**narrow QRS complex**) (**A flutter/ ECG**)

15(O) Alternating between small and big jumps:
Atrial flutter = atria ~300 bpm (“sawtooth” flutter waves) → typically 1 out of every 2 signals filtered through AV node to ventricles (**1:2 block, ventricular rate ~150 bpm**); other ratios (e.g. 1:3, 1:4) also common (**A flutter/ ECG**)

16(O) Dog with flutter ears in front of regular and irregular notes:
Atrial flutter = atria ~300 bpm (“sawtooth” flutter waves) → filtered through AV node to ventricles (**typically no variability in block = regular tachycardia; variable block = irregular tachycardia**) (**A flutter/ ECG**)

17(O) Irregularly irregular TV signal:
Atrial fibrillation = discoordinated pacemaker cells within atrium → **no discernable p waves (fibrillation** = erratic baseline between QRS) → irregular signals filtered through AV node to ventricles (**"irregularly irregular" narrow QRS complex tachycardia ~100-200 bpm**) (**A fib/ ECG**)

18(O) Various "P" tails above "mat":
Multifocal atrial tachycardia (MAT) = signals generated in multiple different areas of the atrium → **at least 3 different P wave morphologies** → each followed by a QRS (**irregular narrow QRS complex tachycardia**) (**MAT/ ECG**)

19(S) Young and old dog:
pSVT (AVRT, AVNRT) can present at **any age** (younger pts are less likely to have underlying structural heart disease) (**pSVT/ Demographics**)

20(S) Disoriented and clutching heart:
pSVT (AVRT, AVNRT) presents with **palpitations, lightheadedness** (chest pain and SOB in patients with underlying heart abnormalities; syncope uncommon) (**pSVT/ Chief complaint**)

21(S) White wolves running down nodal and accessory pathway simultaneously:
history of Wolff Parkinson White indicates the presence of an accessory pathway (AVRT) (**pSVT/ PMHx**)

22(O) White wolf on accessory pathway pulling QRS skirt:
Wolff Parkinson White → **short PR interval + slurred upstroke of QRS** ("delta wave", preexcitation of ventricle by **accessory pathway** before AV node impulse initiates QRS complex) → only seen in some patients with AVRT on resting/nontachy ECG (**pSVT/ ECG**)

23(S) Trigger happy:
pSVT (AVRT, AVNRT) is initiated by a atrial/ventricular premature beat (APB/VPB) → **common triggers** include exercise, alcohol, caffeine, nicotine, stimulants, hypermetabolic disorders (hyperthyroidism) (**pSVT/ Social Hx/PMHx**)

24(O) Doggy ECG bandana:
inspect **ECG** for pSVT (AVRT, AVNRT) → regular, narrow complex tachycardia ~150-250 bpm → no discernable P waves (inspect for retrograde P waves buried inside QRS, inverted, appearing at the wrong time) → may have ST depressions (due to abnormal repolarization) (**pSVT/ ECG**)

25(O) Fainting with BP cuff, brain hat, and floppy heart balloon:
patient with pSVT (AVRT, AVNRT) → assess for **hemodynamic instability** (hypotension, AMS), as well as ischemia/ acute decompensated heart failure (chest pain, diaphoresis, dyspnea, leg swelling, JVD, and crackles) (**pSVT/ Vitals/ Physical Exam**)

26(P) Tazing fainting contestant's dog:
unstable patients with pSVT (AVRT, AVNRT) → **synchronized cardioversion** (all unstable patients with a narrow complex tachyarrhythmia that is not sinus rhythm → cardioversion; unstable sinus tachy → treat underlying cause and consider IV ß-blocker if ischemia/inappropriate tachy) (**pSVT/ Plan (1)**)

27(P) Grabbing neck:
stable patients with pSVT (AVRT, AVNRT) → **vagal maneuvers** (carotid massage, valsalva, ice packs) → IV adenosine (if vagal maneuvers cannot be performed or fail) (**pSVT/ Plan (2)**)

28(P) Dancing dog:
stable patients with pSVT (AVRT, AVNRT) → vagal maneuvers (carotid massage, valsalva, ice packs) → **IV adenosine** (if vagal maneuvers cannot be performed or fail) (**pSVT/ Plan (2)**)

29(P) Exposed "P" tail of dancing dog:
vagal maneuvers and **IV adenosine** can be used to **treat pSVT** (AVRT, AVNRT → terminate reentrant rhythm) OR aid in dx of narrow complex tachy → **identify P waves**(sinus, flutter, retrograde [pSVT], multi morphologies, none [Afib, pSVT]) (**pSVT/ Plan (3)**)

30(P) "Calci-yum" ice cream and muted beta bugler:
stable patients with pSVT (AVRT, AVNRT) → refractory to vagal maneuvers and IV adenosine → can attempt **IV CCB or beta blocker** (or digoxin) → synchronized cardioversion (if refractory to pharmacotherapy) (**pSVT/ Plan (4)**)

31(P) Toward the tazer:
stable patients with pSVT (AVRT, AVNRT) → refractory to vagal maneuvers and IV adenosine → can attempt IV CCB or beta blocker (or digoxin) → **synchronized cardioversion** (if refractory to pharmacotherapy) (**pSVT/ Plan (5)**)

32(P) Chronic clock and ablated radio:
Patients with frequent recurrent episodes can be treated with **long term rate-control agents** (CCB, beta blocker) or **radiofrequency ablation (pSVT/ Plan (6)**)